

SAUMALKOL VOLODARSKOYE, Kazakhstan

WMO: 288770

Lat: 53.317N

Lon: 68.100E

Elev: 319

StdP: 97.55

Time Zone: 6.00 (E06)

Period: 83-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.3	-27.7	-33.7	0.2	-31.2	-30.4	0.2	-27.6	16.5	-13.2	14.4	-7.0	1.9	230	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	30.0	18.3	28.2	18.2	26.3	17.4	20.6	26.6	19.4	25.4	18.5	24.4	3.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.7	23.8	17.2	12.8	22.5	16.2	12.0	21.6	60.7	26.5	56.8	25.3	53.9	24.5	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.0	8.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-33.2	22.0	3.7	1.2	-35.9	22.8	-38.0	23.5	-40.1	24.2	-42.8	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-14.2	-15.3	-8.4	3.1	11.6	18.1	19.1	16.3	10.6	3.1	-8.7	-12.4
	DBStd	13.74	6.30	6.09	5.41	6.20	5.50	5.05	3.66	3.45	4.72	5.03	6.73	6.24
	HDD10.0	3831	749	709	571	223	48	6	0	1	48	219	561	695
	HDD18.3	6105	1007	943	829	458	214	64	36	82	234	472	811	953
	CDD10.0	910	0	0	0	15	99	248	281	196	66	5	0	0
	CDD18.3	144	0	0	0	0	7	57	59	19	2	0	0	0
	CDH23.3	1255	0	0	0	1	82	565	457	122	28	0	0	0
	CDH26.7	334	0	0	0	0	10	179	126	18	1	0	0	0
Wind	WSAvg	3.3	4.3	4.0	3.0	3.5	3.2	2.6	2.4	2.4	3.0	3.2	3.4	4.1
Precipitation	PrecAvg	426	24	18	22	27	32	44	75	51	32	35	33	29
	PrecMax	552	48	37	50	92	54	83	157	125	63	59	63	58
	PrecMin	266	7	3	5	0	3	13	29	7	11	5	6	13
	PrecStd	85	9	9	13	23	16	20	41	38	15	16	14	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	-1.3	3.7	22.0	27.9	33.1	32.4	29.0	26.4	19.3	4.7	1.6
		MCWB	-0.8	-2.0	1.0	11.8	15.5	17.9	21.1	19.1	15.2	10.4	3.0	0.6
	2%	DB	-1.3	-2.6	1.7	18.9	25.6	30.7	29.6	26.7	23.7	14.9	1.7	-0.9
		MCWB	-1.9	-3.2	0.1	10.7	14.5	18.1	19.8	18.2	14.6	8.3	0.1	-1.3
	5%	DB	-3.4	-4.3	0.4	15.9	23.4	28.9	27.8	24.7	20.7	12.5	0.6	-2.4
		MCWB	-4.0	-5.1	-0.7	9.3	13.4	17.7	19.2	17.2	12.9	7.0	-0.4	-3.1
	10%	DB	-5.1	-6.3	-0.8	12.5	21.0	26.8	25.9	22.8	18.3	10.4	-0.6	-3.9
		MCWB	-5.8	-7.2	-1.9	7.9	12.7	17.2	18.5	16.7	11.9	6.2	-1.4	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	-1.9	2.0	12.8	16.7	21.1	22.6	20.9	18.2	11.4	3.1	0.9
		MCDB	-0.1	-1.6	3.2	19.2	24.9	29.0	29.8	25.2	24.1	17.1	4.9	1.2
	2%	WB	-1.9	-3.2	0.4	11.2	15.7	19.6	21.2	19.1	15.1	9.5	0.7	-1.2
		MCDB	-1.2	-2.7	1.3	17.6	23.4	27.2	27.5	24.3	21.4	14.0	1.2	-0.9
	5%	WB	-4.0	-5.0	-0.7	9.7	14.3	18.6	20.1	18.2	13.5	7.7	-0.4	-3.0
		MCDB	-3.5	-4.3	0.3	15.8	21.3	26.6	25.7	22.9	19.2	11.6	0.3	-2.5
	10%	WB	-5.7	-7.1	-1.9	7.9	13.2	17.6	19.1	17.2	12.2	6.3	-1.5	-4.4
		MCDB	-5.1	-6.2	-0.9	12.3	20.1	25.4	24.7	21.9	17.6	10.1	-0.7	-3.9
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.7	8.3	8.6	10.8	10.8	9.2	8.9	9.6	7.9	7.1	7.2
		MCDBR	7.9	6.4	7.9	12.6	14.2	13.2	11.9	12.2	13.9	11.6	7.0	8.3
		MCWBR	7.8	6.1	6.8	6.8	6.8	5.8	5.0	5.7	7.1	6.4	6.5	7.9
	5% WB	MCDBR	7.3	6.3	7.3	12.4	12.6	11.8	10.6	10.6	12.3	10.7	7.2	7.8
		MCWBR	7.3	6.2	6.6	7.5	6.7	6.2	5.3	5.7	7.5	6.6	6.9	7.8
Clear-Sky Solar Irradiance	taub	0.281	0.311	0.365	0.401	0.399	0.400	0.420	0.416	0.365	0.345	0.315	0.281	
	taud	2.076	2.042	1.984	2.143	2.235	2.309	2.243	2.236	2.335	2.278	2.140	2.057	
	Ebn at Noon	662	762	794	824	849	851	824	802	804	728	610	573	
	Edh at Noon	72	104	138	135	130	122	128	122	97	83	68	61	
All-Sky Solar Radiation	RadAvg	1.17	2.28	3.76	4.64	5.68	6.07	5.60	4.75	3.43	1.81	0.89	0.78	
	RadStd	0.08	0.10	0.40	0.49	0.47	0.49	0.44	0.41	0.25	0.27	0.13	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page